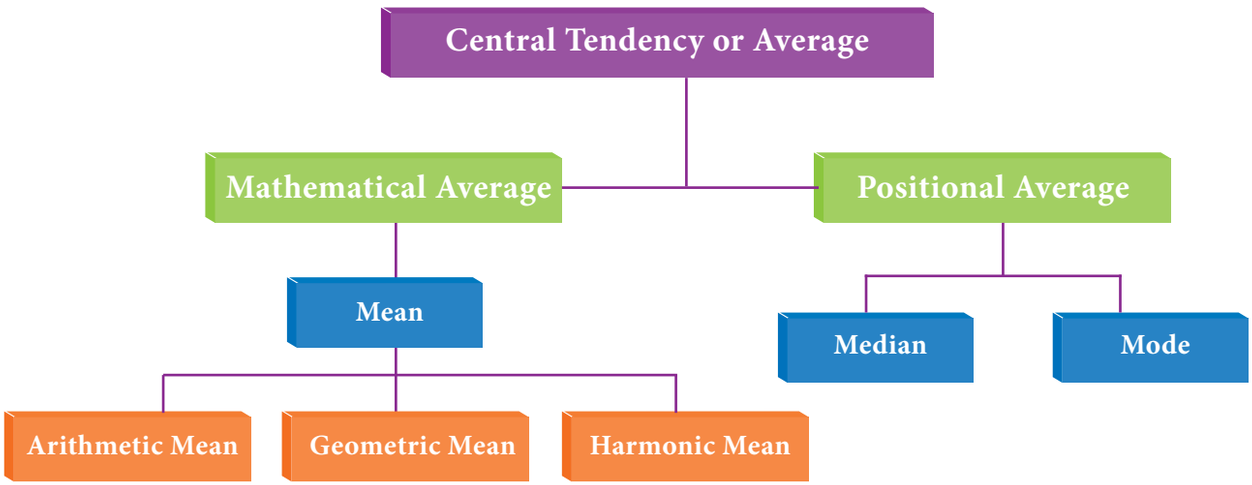
- It should be amenable for further mathematical calculations.
- It should be least affected by the fluctuations of the sampling.
- It should not be unduly affected by the extreme values.

5.3 Various measures of central tendency



5.3.1 Arithmetic Mean

(a) To find A.M. for Raw data

For a raw data, the arithmetic mean of a series of numbers is sum of all observations divided by the number of observations in the series. Thus if $x_1, x_2, ..., x_n$ represent the values of n observations, then arithmetic mean (A.M.) for n observations is: (direct method)

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

There are two methods for computing the A.M :

(i) Direct method (ii) Short cut method.

NOTE

Normal health parameters such as blood pressure, pulse rate, blood cell count, BMI, blood sugar level etc., are calculated averages of people in a particular region and always vary among individuals.

Example 5.1

The following data represent the number of books issued in a school library on selected from 7 different days 7, 9, 12, 15, 5, 4, 11 find the mean number of books.

Solution:

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

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